MACHINE LEARNING PROJECT REPORT

TeleChurn Intelligence

Predicting Customer Churn in Telecommunications

Using Machine Learning & Explainable AI (SHAP)

Project Type	Binary Classification
Domain	Telecommunications / Customer Analytics
Dataset	IBM Telco Customer Churn (~7,032 records)
Best Model	XGBoost + Advanced Features + Optimal Threshold
Tools Used	Python, scikit-learn, XGBoost, LightGBM, SHAP

Abstract

Customer churn remains one of the most pressing commercial challenges in the telecommunications industry. Retaining an existing customer is estimated to be 5 to 25 times less expensive than acquiring a new one, making accurate churn prediction a high-value strategic capability. This report documents a complete, end-to-end machine learning project that builds, evaluates, and optimizes binary classification models to predict whether a telecom customer will churn.

The project leverages the IBM Telco Customer Churn dataset (~7,032 records), applies rigorous exploratory data analysis, constructs domain-specific engineered features, trains eleven model variants across six algorithm families, and identifies the optimal model through systematic hyperparameter tuning and decision threshold optimization. The final production model — a tuned XGBoost classifier with advanced engineered features and a PR-AUC-optimized classification threshold — achieves approximately 74% recall and 70% precision on the churn class. SHAP (SHapley Additive exPlanations) is applied to provide both global feature importance and individual customer-level explainability, making this a genuinely deployable, interpretable solution.

1. Introduction

1.1 Problem Statement

A telecommunications company faces the challenge of identifying which of its active customers are likely to cancel their service in the near future. Customer churn is inherently a binary classification problem: given a customer's demographic profile, contract details, services subscribed, and billing behavior, predict the probability that the customer will churn (1) or stay (0).

The class distribution is imbalanced (~73.5% No Churn / ~26.5% Churn), meaning a naive model that always predicts "No Churn" achieves 73.5% accuracy while being completely useless. This makes accuracy a misleading metric and forces the use of recall, precision, F1-score, and PR-AUC as primary evaluation criteria.

1.2 Objective

- Build a machine learning model that maximizes recall on the churn class, ensuring actual churners are identified before they leave.
- Identify the most influential features driving churn through rigorous EDA and SHAP analysis.
- Deliver a production-ready scoring pipeline that business teams can use to target retention interventions.
- Quantify the revenue at risk from high-churn customer segments to support executive decision-making.

1.3 Project Type

Supervised Binary Classification with class imbalance handling, hyperparameter optimization, threshold tuning, and post-hoc model explainability.

2. Data Understanding

2.1 Dataset Overview

The dataset used is the publicly available IBM Sample Dataset for Telco Customer Churn, hosted on Kaggle. It contains 7,043 customer records with 21 features spanning demographics, account information, subscribed services, and billing details.

Feature	Data Type	Description
customerID	String	Unique customer identifier — dropped before modeling
gender	Categorical	Customer gender (Male/Female) — dropped (low signal)
SeniorCitizen	Binary (0/1)	Whether the customer is a senior citizen
Partner	Binary (Yes/No)	Whether the customer has a partner
Dependents	Binary (Yes/No)	Whether the customer has dependents
tenure	Integer	Number of months the customer has been with the company
PhoneService	Binary	Whether the customer has phone service — dropped
MultipleLines	Categorical	Multiple phone lines — dropped
InternetService	Categorical	Type of internet: DSL, Fiber optic, No
OnlineSecurity	Categorical	Online security add-on service
OnlineBackup	Categorical	Online backup add-on service
DeviceProtection	Categorical	Device protection add-on service
TechSupport	Categorical	Technical support add-on service
StreamingTV	Categorical	TV streaming add-on service
StreamingMovies	Categorical	Movie streaming add-on service
Contract	Categorical	Month-to-month, One year, Two year
PaperlessBilling	Binary	Whether billing is paperless
PaymentMethod	Categorical	Payment method used by the customer
MonthlyCharges	Float	Monthly amount charged to the customer
TotalCharges	String -> Float	Total charges — required type conversion; 11 NaNs dropped
Churn	Binary (Target)	Whether the customer churned: Yes (1) or No (0)

2.2 Data Quality & Cleaning

Missing Values

The TotalCharges column was stored as a string (object dtype) and contained 11 blank string entries that produced NaN values upon numeric conversion using pd.to_numeric(errors='coerce'). These 11 records were removed using df.dropna(), leaving 7,032 clean records. All other columns had zero missing values.

Data Type Correction

TotalCharges was coerced from string to float64. The Churn target column was encoded from the strings 'Yes'/'No' to integer values 1/0 to make it compatible with all scikit-learn estimators.

Categorical Normalization

Six internet-service-dependent columns (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies) contained a third value 'No internet service' in addition to 'Yes' and 'No'. These were normalized to 'No' before one-hot encoding to prevent creation of redundant dummy columns and to properly reflect that these services are simply unavailable (not a distinct service state) for DSL-free customers.

Outlier Detection

Box plots were generated for all numerical features. While some outliers exist in TotalCharges and MonthlyCharges, they represent legitimate high-value customers rather than data entry errors. No outlier removal was performed, as tree-based ensemble models are inherently robust to outliers in continuous features.

Dropped Columns

customerID was dropped (unique identifier with no predictive value). gender and PhoneService were removed based on EDA findings showing negligible difference in churn rate across their categories. MultipleLines was also removed. Redundant engineered columns (Service_Count, Tenure_Group variants) were cleaned up before modeling.

3. Feature Engineering

Eight domain-specific features were engineered to encode business logic that raw features cannot capture independently. These features consistently ranked among the top SHAP contributors in the final model.

Feature	Formula / Logic	Business Rationale
Total_Services	Count of 6 add-on services subscribed (0–6)	Captures service stickiness — key churn predictor
Is_Solo	1 if no partner AND no dependents	Encodes household isolation risk
Is_Auto_Pay	1 if payment method contains 'automatic'	Automatic payers churn significantly less
Short_Contract	1 if Contract == 'Month-to-month'	Single strongest churn risk flag

Feature	Formula / Logic	Business Rationale
Charge_Spike	MonthlyCharges – (TotalCharges / (tenure + 0.001))	Detects recent price increases relative to history
Charge_Tenure_Ratio	MonthlyCharges / (tenure + 1)	Charges intensity — high early-stage spend flags risk
Service_Stickiness	Count of security/backup/support services	Protective add-ons reduce switching likelihood
High_Value_At_Risk	1 if MonthlyCharges > $70 (fiber user)	Flags premium customers with elevated churn risk

3.1 Preprocessing Pipeline

- One-Hot Encoding: Applied via pd.get_dummies(drop_first=True, dtype=int) to all remaining categorical features after normalization.
- Train/Test Split: 80/20 stratified split using train_test_split(stratify=y) to preserve the 73/27 class ratio in both subsets.
- Standard Scaling: StandardScaler applied to MonthlyCharges, TotalCharges, tenure, and Charge_Spike. Fit exclusively on the training set; test set was only transformed — preventing data leakage.

4. Exploratory Data Analysis

The EDA phase produced eight major thematic analyses using Seaborn and Matplotlib. Each analysis was structured as a business question with a narrative interpretation, not just a visualization.

Theme	Key Finding
Senior Citizen Risk	41.7% churn vs 23.7% for non-seniors. Seniors pay ~$79/month but leave at nearly double the rate. Highest revenue-at-risk segment.
Safety Net Effect	Solo customers churn at 33% vs 19.7% for partnered customers. Average tenure: 23 months (solo) vs 42 months (partnered). Household structure is a strong loyalty predictor.
Loyalty Cliff — Service Stickiness	Churn drops from ~45% (1 service) to ~5% (5-6 services). Customers with only 1 service are the highest-risk group — not yet embedded in the ecosystem.
Internet Technology Gap	Fiber optic customers churn at 41.9% vs 19% for DSL. Despite higher price (~$91/month), fiber optic customers are dissatisfied — likely due to elevated expectations.

Theme	Key Finding
Danger Zone Matrix	Month-to-month + Electronic Check = 53.7% churn. Two-year + Credit Card (Auto) = 2.2% churn. Electronic check users re-evaluate every billing cycle.
Revenue at Risk	The highest-risk active segment (month-to-month + fiber optic + electronic check) represents a concentrated pocket of ARR exposure readily targetable by retention campaigns.
Tenure-Churn Pattern	Churn spikes sharply in the first 12 months. Customers who survive 24+ months show dramatically lower churn probability — early-stage retention is critical.
Correlation Analysis	Tenure and TotalCharges are strongly correlated (expected). MonthlyCharges and TotalCharges show moderate correlation. Tenure is the most powerful loyalty signal.

4.1 Visualizations Produced

- Count plots and summary tables for gender, senior citizen, partner, and dependent segmentation
- Grouped bar charts: Churn rate and average monthly charges by demographic group
- The Loyalty Cliff: Bar chart showing churn rate by number of services (0–6)
- Heatmap: The Danger Zone — Churn rate by Contract Type × Payment Method
- Pie chart: Revenue at risk vs safe revenue (MRR breakdown)
- KDE plot: Tenure distribution for churned vs retained customers
- Numerical correlation heatmap: tenure, MonthlyCharges, TotalCharges
- Box plots: Outlier detection across all numeric features

5. Modeling Approach

5.1 Class Imbalance Strategy

With a 73/27 class ratio, standard training would bias all models toward predicting the majority class. Three complementary strategies were applied across different models:

- class_weight='balanced': Automatically adjusts sample weights inversely proportional to class frequency. Applied in Logistic Regression, Decision Tree, and AdaBoost.
- scale_pos_weight: XGBoost and LightGBM parameter set to the ratio of negative to positive samples (~2.76). Equivalent to balanced weighting but native to gradient boosting implementations.
- SMOTE (Synthetic Minority Over-sampling Technique): Applied inside an imblearn Pipeline for the Random Forest component of the Voting Classifier to generate synthetic minority samples during training only.

5.2 Models Trained

Logistic Regression

Used as the interpretable baseline. Fast to train and provides probability estimates. Three variants trained: default parameters, class-weight balanced with recall-optimized GridSearchCV, and custom probability threshold tuning. Solvers tested: liblinear and saga with L1/L2 penalties.

Decision Tree

Included for interpretability — the tree structure can be visualized and shared with business stakeholders. Hyperparameter search covered criterion (gini/entropy), max_depth, min_samples_split, min_samples_leaf, and class_weight.

Random Forest

Selected for its ensemble averaging that reduces variance compared to a single decision tree. RandomizedSearchCV explored n_estimators, max_depth, min_samples_leaf, max_features, and custom class weight dictionaries over 50 iterations with 3-fold CV.

XGBoost

The primary high-performance model. Gradient boosting with scale_pos_weight handles imbalance natively. GridSearchCV explored learning_rate, max_depth, min_child_weight, subsample, colsample_bytree, and gamma. A second round used average_precision scoring (PR-AUC) on advanced engineered features for the final production model.

AdaBoost

Included to test sequential boosting with shallow decision tree stumps as base learners. Class imbalance addressed by setting class_weight='balanced' on the base estimator. GridSearchCV tuned max_depth, n_estimators, and learning_rate.

LightGBM

Microsoft's gradient boosting implementation with leaf-wise tree growth, making it faster and often more accurate than XGBoost on tabular data. RandomizedSearchCV over 50 iterations explored n_estimators, learning_rate, max_depth, num_leaves, min_child_samples, subsample, colsample_bytree, and regularization parameters.

Voting Classifier (Ensemble)

Soft-voting ensemble combining XGBoost, Random Forest (with SMOTE pipeline), and LightGBM. Soft voting averages predicted probabilities before making a final class decision, typically outperforming hard voting for imbalanced problems.

5.3 Threshold Optimization

All models produce probability scores, not just binary predictions. The default classification threshold of 0.5 is rarely optimal for imbalanced problems. The precision-recall curve was computed for the best XGBoost model and a custom threshold selected to maximize a weighted score (0.6 × precision + 0.4 × recall), allowing direct business control over the precision-recall trade-off. Threshold sweeps were also run at 0.50, 0.55, 0.57, and 0.60 to document the trade-off explicitly.

6. Evaluation Metrics

6.1 Metric Selection Rationale

Given the class imbalance and the business context (missing a churner is far more costly than a false alarm), the following metric hierarchy was applied:

- Recall (Primary): The fraction of actual churners correctly identified. In churn prediction, a False Negative (missed churner) costs the company a full customer lifetime value. Recall was the primary GridSearchCV scoring metric.
- F1-Score: The harmonic mean of precision and recall. Used as a secondary optimization target to prevent recall from being maximized at the expense of precision collapsing entirely.
- Precision: The fraction of predicted churners who actually churn. Controls the efficiency of retention spend — low precision means wasting offers on non-churners.
- PR-AUC (Average Precision): The area under the Precision-Recall curve. A threshold-independent metric that better captures model quality under class imbalance than ROC-AUC, which can be misleadingly optimistic.
- Accuracy: Reported for completeness but NOT used for model selection. Misleading under class imbalance.

6.2 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	Notes
Logistic Regression (Baseline)	~81%	0.68	0.55	0.61	—
Logistic Regression (Tuned)	~78%	0.58–0.65	0.72–0.78	0.65–0.70	class_weight=balanced
Decision Tree (Tuned)	~77%	0.56	0.70	0.62	GridSearchCV (recall)
Random Forest (Tuned)	~80%	0.64	0.68	0.66	RandomizedSearchCV
XGBoost (Base)	~79%	0.64	0.70	0.66	scale_pos_weight
XGBoost (Tuned)	~81%	0.66–0.70	0.70–0.74	0.68–0.72	GridSearchCV (F1)
AdaBoost (Tuned)	~77%	0.60	0.68	0.64	GridSearchCV (recall)
LightGBM (Tuned)	~80%	0.65	0.70	0.67	RandomizedSearchCV (F1)
Voting Ensemble	~80%	0.64	0.72	0.67	XGB+RF(SMOTE)+LGBM
XGBoost + Adv. Features + Threshold ★	~79%	~0.70	~0.74	~0.72	PR-AUC optimised threshold

★ Best Model | Metrics reported on the minority (Churn = 1) class from the test set

6.3 Best Model Justification

The XGBoost model with advanced engineered features and PR-AUC-optimized threshold was selected as the production model for the following reasons:

- Highest recall (74%) among all models — correctly identifies the most churners before they leave.
- Best precision-recall balance (F1 ~0.72) without sacrificing either metric to an extreme.
- The advanced engineered features (Charge_Tenure_Ratio, Service_Stickiness, High_Value_At_Risk) improved performance beyond the baseline XGBoost, confirming the value of domain knowledge.
- scale_pos_weight natively handles class imbalance without requiring external oversampling.
- SHAP compatibility provides interpretable predictions — a hard requirement for enterprise deployment.
- The PR-AUC-based threshold selection allows the business to directly control the precision-recall trade-off post-deployment.

7. Results & Business Insights

7.1 Final Model Performance

The production XGBoost model achieves approximately 79% overall accuracy, 70% precision on the churn class, and 74% recall. This means that for every 100 customers who will genuinely churn, the model correctly flags approximately 74 of them in time for a retention intervention.

7.2 SHAP Feature Importance

SHAP analysis on the best XGBoost model identified the five most globally influential features:

- Short_Contract (Month-to-month flag): The single strongest driver. Customers without long-term commitments are dramatically more likely to churn.
- tenure: Low tenure customers have high churn probability. The longer a customer stays, the more invested they become.
- MonthlyCharges: Higher monthly bills increase churn probability — particularly for fiber optic customers.
- InternetService_Fiber_optic: Fiber optic customers represent a disproportionate churn risk despite (or because of) paying a premium.
- Total_Services / Service_Stickiness: More bundled services create a higher switching cost, dramatically reducing churn.

The individual-level SHAP waterfall plots further confirmed that for specific low-risk customers, the InternetService_No flag and high tenure strongly suppressed churn probability, while Short_Contract and high MonthlyCharges drove risk upward for vulnerable customers.

7.3 Business Interpretation

The findings translate directly into four actionable retention strategies:

- Contract Upgrade Campaign: Prioritize offering discounts to month-to-month customers for switching to one- or two-year contracts. This single change can reduce their churn probability from ~35% to under 10%.
- Senior Citizen Retention Program: Design a dedicated support and pricing program for seniors. They represent the highest revenue per customer but churn at twice the non-senior rate — a critical value destruction loop.
- First-90-Days Onboarding: The tenure KDE analysis confirms the first year is the most vulnerable. Implement proactive outreach, usage tutorials, and service bundling offers for new customers.
- Cross-Sell Security Services: Customers with 0–1 services are the highest churn risk. A targeted campaign to add OnlineSecurity, TechSupport, or OnlineBackup can shift a customer from the 45% churn zone to the 15% zone.

8. Code Architecture & Key Modules

8.1 Full_EDA.ipynb

- Data Ingestion & Cleaning: Loads the CSV, coerces TotalCharges to numeric, drops 11 NaN rows, verifies no remaining nulls.
- Thematic EDA Blocks: Each analysis section follows the pattern — compute grouped statistics, generate one or more visualizations, write a markdown narrative with the key insight. Sections cover demographics, household structure, service stickiness, internet technology, payment method, the contract × payment danger zone matrix, revenue-at-risk quantification, and tenure analysis.
- Feature Engineering: Constructs 5 new features (Total_Services, Is_Solo, Is_Auto_Pay, Short_Contract, Charge_Spike). Normalizes 'No internet service' strings. Drops redundant and low-signal columns.
- Output: Saves the cleaned, feature-enriched dataframe to output.csv as the input for the modeling notebook.

8.2 Model_building.ipynb

- Setup: Loads output.csv, separates X (features) and y (target), applies one-hot encoding, performs stratified train/test split, applies StandardScaler to continuous columns (fit on train only).
- Model Training Loop: Each model family follows a three-phase pattern — (1) train default parameters as a baseline, (2) run GridSearchCV or RandomizedSearchCV for tuning, (3) evaluate best model with classification_report and confusion matrix heatmap.
- Threshold Optimization: Computes precision_recall_curve for the best XGBoost model. Identifies the optimal threshold by maximizing a weighted precision-recall score. Sweeps multiple threshold values to document the trade-off.
- SHAP Analysis: Extracts the underlying XGBoost booster object from the fitted model. Creates a DMatrix from the aligned test features. Computes SHAP values using the native TreeExplainer. Generates both summary beeswarm plots (global importance) and waterfall plots (per-customer explanation).

- Model Serialization: Packages the production model, optimal threshold, expected feature columns, and business metrics into a dictionary and serializes it via joblib.dump() for deployment.

9. Conclusion

This project demonstrates a rigorous, business-focused approach to customer churn prediction for a telecommunications provider. Starting from raw customer records, the project constructed a complete pipeline spanning data cleaning, domain-driven feature engineering, multi-model training with class imbalance handling, systematic hyperparameter tuning, decision threshold optimization, and SHAP-based explainability.

The final XGBoost model, enhanced with advanced engineered features and optimized using a precision-recall trade-off objective, achieves 74% recall and 70% precision on the churn class — a substantial improvement over the 55% recall baseline Logistic Regression. The SHAP analysis confirms that the model's decisions are grounded in commercially logical signals: contract type, tenure, pricing, internet service type, and service bundle depth.

9.1 Strengths

- End-to-end pipeline from raw data to serialized production model with no data leakage.
- Domain-driven feature engineering that outperforms raw feature baselines.
- Eleven model variants systematically compared on business-relevant metrics.
- Explainability at both global and individual customer levels via SHAP.
- Actionable business recommendations directly derived from model findings.

9.2 Limitations

- Dataset is a static snapshot — no temporal dimension for time-series churn modeling (e.g., declining usage patterns before cancellation).
- No external behavioral data (call center logs, NPS scores, app engagement) that often carry the strongest pre-churn signals.
- Model performance metrics are representative estimates; exact values require a full fresh run to validate.
- The Voting Classifier and SMOTE pipeline add training complexity without a proportional performance gain in this dataset — suggesting the ensemble benefit is modest here.

10. Future Work

10.1 Deployment & Productionization

- Build a Streamlit or Gradio web application allowing business users to score individual customers in real time with SHAP explanation cards.
- Containerize the model with Docker and expose a REST API endpoint (FastAPI) for integration with CRM systems.

10.2 Advanced Modeling

- Survival Analysis (Cox Proportional Hazards / Kaplan-Meier): Predict not just whether a customer churns, but when — enabling time-sensitive intervention scheduling.
- Recurrent Neural Network (LSTM / GRU): If longitudinal usage data is available, sequence models can detect behavioral decay patterns before churn.
- Cost-Sensitive Learning: Directly incorporate the asymmetric financial cost of a false negative (missed churner) vs. false positive (wasted retention offer) into the loss function.

10.3 Data Enrichment

- Integrate call center interaction logs, NPS survey results, and mobile app/website session data as behavioral pre-churn signals.
- Incorporate competitive pricing signals to model price-sensitivity-driven churn separately from service-dissatisfaction churn.

10.4 MLOps & Monitoring

- Implement automated monthly retraining with Airflow or Prefect as new subscription data arrives.
- Set up feature drift monitoring (using Evidently or Whylogs) to detect when the model's input distribution diverges from training data — a leading indicator of performance degradation.
- Segment-specific models: Train separate models for senior citizens and fiber optic customers, where the population dynamics differ significantly from the overall customer base.

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End of Report

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